

VOC

A PERSONAL MEMORY LAB

VOC learns how you forget.

A vocabulary and grammar app that builds a personal model of your memory — and uses it to decide what you should review, when, and why.

LEARNER → VOCABULARY → MEMORY → PREDICTION

VOC is a vocabulary and grammar learning app with one key idea: instead of giving everyone the same review schedule, it learns how YOUR memory works. For each word, it predicts when you are about to forget it and why — and then gives you the right practice at the right moment. The chain at the bottom is the whole product in four words: the learner studies vocabulary, VOC models their memory, and uses that model to make predictions.

Learning a word does not mean you will remember it.

Forgetting is not random — it is personal. Each learner loses different words at a different speed.



Day 1

Newly learned words feel fresh and easy to recall.



Day 3

Some words start to fade, and recall gets slower.



Day 7

Many words are gone unless they were reviewed.

The question VOC answers: which of *your* words will you forget, when — and why?

Everyone who learns a language knows this: you learn 100 words, and a week later many of them are gone. This is the forgetting curve. On day 1 the words are fresh. By day 3 some start to fade. By day 7, the words you did not review are mostly lost. The important point is that this is personal: two learners forget different words, at different speeds, for different reasons. So the real question is not "did you learn this word?" but "which of YOUR words will you forget, when, and why?" That is the question VOC is built to answer.

A fixed schedule treats every learner the same

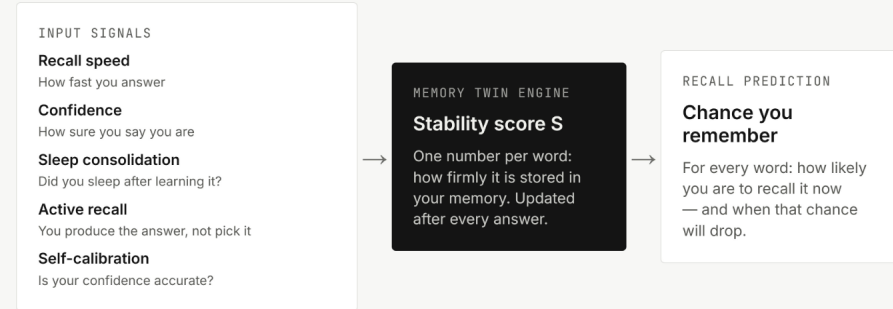
VOC replaces "review on day 1, 3, 7" with a prediction made for you and for each word.

	FIXED REVIEW SCHEDULE	VOC
When to review	Same intervals for everyone	Predicted for you, word by word
What it learns from you	Right or wrong	Speed, confidence, accuracy, sleep, history
After a mistake	Shows the same card again	Finds the likely cause and picks a fix
Result	Generic practice	Practice aimed at your weakest memories

Many learning tools use a fixed review rhythm — for example, review a word after 1 day, then 3 days, then 7 days. That is better than nothing, but it assumes all learners and all words behave the same. VOC is different in four ways: it predicts the right review time for each person and each word; it learns from more than right/wrong — how fast you answer, how confident you are, and more; when you make a mistake it tries to find out why; and as a result, your practice time goes to the words you are actually about to lose.

How VOC measures your memory

Signals from every answer go in. A stability score for each word comes out. That score predicts recall.



This is the core of VOC, in three steps. First, the inputs: every time you answer, VOC records signals — how fast you answered, how confident you felt, whether you had a night of sleep since learning the word (sleep helps memory consolidate), whether you actively produced the answer instead of choosing from options, and whether your confidence matches your real accuracy. Second, the Memory Twin Engine combines these signals into one number per word: its stability, S . A high S means the word is firmly stored. Third, from S , VOC predicts the probability that you can still recall the word right now, and how that probability will fall over time. The next slide shows the formula behind that prediction.

The memory formula, in plain words

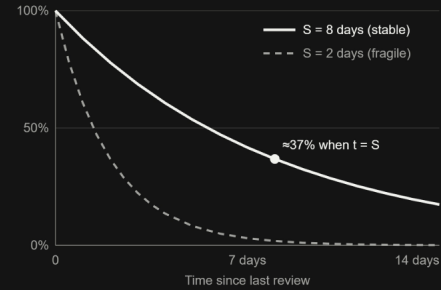
$$P(t) = e^{-t/S}$$

$P(t)$ chance you still recall the word

t days since your last review

S stability: how slowly you forget

Example: if $S = 5$ days, then 5 days after a review $P \approx 37\%$. VOC reviews the word *before* P gets too low.

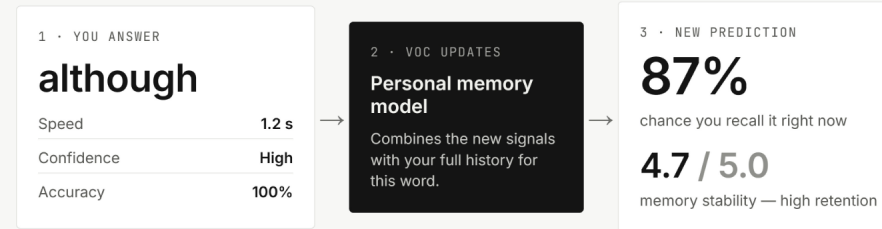


The prediction uses a simple, well-known forgetting-curve formula: P of t equals e to the minus t over S . $P(t)$ is the probability you can still recall the word. t is the time since you last reviewed it. S is the stability of that memory. The bigger S is, the slower the curve falls. A useful rule of thumb: when the time since review equals S , the recall chance has dropped to about 37%. The chart compares two words: a stable one with $S = 8$ days, which you still have a good chance of remembering after a week, and a fragile one with $S = 2$ days, which is almost gone within a week. VOC's job is to estimate S for every word, for every learner, and to schedule the review just before the chance of remembering gets too low.

02 · ONE ANSWER, STEP BY STEP

How VOC learns you

Every attempt becomes a learning signal. The more you use VOC, the better it knows your memory.



VOC — Personal Memory Lab · example values

06

Here is one real interaction, step by step. Step 1: the learner is asked for the word "although" and answers correctly in 1.2 seconds, with high confidence. Step 2: VOC's personal memory model takes these signals and combines them with everything it already knows about this learner and this word. Step 3: it produces an updated prediction — an 87% chance the learner can recall the word right now, and a stability of 4.7 out of 5, meaning the memory is strong. Because the answer was fast, confident and correct, the next review of this word can wait longer. These numbers are an example to show how it works.

Predictive diagnostics: four steps after a mistake

01

Detect

VOC notices a failed or very slow recall.

02

Diagnose

It ranks the most likely reasons you forgot.

03

Intervene

It picks the drill that fixes that exact reason.

04

Re-forecast

It updates the memory curve and the next review date.

A schedule only knows *that* you forgot. **VOC tries to find out why.**

This is what makes VOC more than a review scheduler. When you get a word wrong, VOC runs four steps. Detect: it notices the failed or unusually slow answer. Diagnose: it looks at your history and estimates the most likely causes — for example, you confused it with a similar word, or too much time passed since the last review. Intervene: it chooses a practice type that targets that specific cause, instead of just showing the same card again. Re-forecast: it recalculates the memory curve for the word and sets a new review date. The next slide walks through a concrete example.

The learner forgets "although"

DIAGNOSE · LIKELY CAUSES

Confused with a similar word	42%
Review gap too long	31%
Weak confidence history	18%
Too little exposure	9%

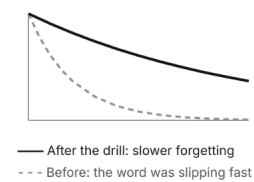
INTERVENE · THE FIX

Contrastive drill

Practise "although" next to the word it gets mixed up with, until the difference is automatic.

He decided to go ___ the heavy rain.
despite ✓ + noun · **although**
+ clause

RE-FORECAST



A concrete example. The learner fails to recall "although". VOC's diagnosis says the most likely cause, at 42%, is confusion with a similar word; then a review gap that was too long, 31%; weak confidence history, 18%; and too little exposure, 9%. Because confusion is the top cause, VOC chooses a contrastive drill: the learner practises "although" side by side with the word it gets confused with — here, "despite". For example: "He decided to go ___ the heavy rain." The answer is "despite", because "despite" is followed by a noun, while "although" is followed by a full clause. Finally VOC re-forecasts: the dashed line shows how fast the word was being forgotten before; the solid line shows the slower forgetting expected after the drill. The percentages are example values.

Three modules, one memory engine

01 · CORE

Daily practice

Shows your weakest memories first, so each session starts where forgetting is closest.

although
recall 24% · S 1.2 days

02 · DRILLS

Grammar & tests

Short adaptive tests check grammar in context and find the rules you mix up.

Conjunctions & clauses
78% mastered · test 3/5

03 · INSIGHTS

Insight & analysis

Tracks progress and flags "leech" words — the ones you keep forgetting.

Retention +14% this week
leech: although, nonetheless

MEMORY ENGINE

One real-time model of your memory, shared by all three modules

The product has three modules, all running on the same memory engine. Daily Practice is the core: it builds your review queue from the words you are closest to forgetting — for example, "although" with only a 24% recall chance and a stability of 1.2 days goes to the front of the line. Grammar and Tests are diagnostic drills: short adaptive tests that check grammar in real sentences and find which rules you confuse. Insight and Analysis shows your progress — retention trends, average stability, recall rate — and flags "leech" words, the ones you keep forgetting again and again. Because all three share one memory engine, what you do in one module improves the predictions in the others.

Every attempt makes both sides better

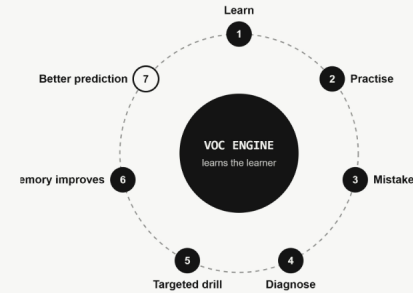
The learner improves

Steps 1-6: learn, practise, make a mistake, get a diagnosis and a targeted drill — and remember better.

VOC improves

Step 7: every answer, including every mistake, makes the next prediction for this learner more accurate.

Every mistake teaches the learner.
Every attempt teaches VOC.



This loop is why VOC gets better over time. Follow the circle: you learn a word, practise it, at some point you make a mistake, VOC diagnoses why, gives you a targeted drill, and your memory of the word improves. That is the learner's side. At the same time, step 7: every answer — especially every mistake — gives VOC more data about how this specific learner forgets, so its next prediction is more accurate. Then the loop starts again. Two things improve together: the learner's memory and VOC's understanding of that learner.

Free for learners, paid by institutions

Free individual use brings learners in. Schools and organisations pay for the insight on top.

01 · LEARNERS

Free

Anyone can use VOC's memory tools at no cost. No barrier means wide adoption — and more data to improve the model.



02 · PLATFORM

VOC engine

- Personal memory model
- Vocabulary & grammar
- Recall probability
- Learning insights



03 · INSTITUTIONS

Paid licence

Schools, universities and government programmes pay for dashboards, progress tracking and group analytics.

The business model has two sides. Individual learners use VOC for free. This removes any barrier to adoption, and every learner also helps the memory model get better. The money comes from institutions: schools, universities and government education programmes buy a paid licence. What they get is the institutional layer on top of the same engine — an admin dashboard, progress tracking for each student, and analytics for whole groups and classes. So: free for learners, revenue from institutional licensing.

Start with learners, then sell to institutions

Free learners prove the model works. Pilot data becomes the sales case for paid licences.

PHASE 1 • B2C

Win individual learners

Who: students preparing for IELTS, CEFR and university exams — they need many words, fast.

How: free app, short-video content, Telegram, referrals.



PHASE 2 • PILOTS

Pilot with learning centres

Who: private language and exam-prep centres.

How: a free one-semester pilot; teachers get a class dashboard; we measure retention.



PHASE 3 • B2B

Sell institutional licences

Who: schools, universities, government programmes.

How: pilot results become case studies; licence per student or per institution.

Our go-to-market runs in three phases, and the order matters. Phase 1: we start with individual learners, because they are the easiest to reach and the app is free for them. Our first focus is students preparing for exams like IELTS, CEFR and university entrance — they need to learn a lot of vocabulary quickly, so they feel the forgetting problem most. Phase 2: we run free, one-semester pilots with private language and exam-prep centres. Teachers get a class dashboard, and we measure how much more students retain. Phase 3: we use those pilot results as case studies to sell paid licences to schools, universities and government education programmes. In short: free learners prove the model works, and pilot data becomes our sales argument.

How we get our first users and customers

Low-cost, organic channels first. Paid ads only once we know what converts.

LEARNERS · B2C

Short-video content

Daily memory tips on Reels, TikTok and Shorts

Telegram channel and bot

Word of the day plus a quiz bot linked to the app

Referrals

Invite a friend, both unlock extra drill packs

Shareable memory report

Weekly progress card learners like to share

INSTITUTIONS · B2B

Free pilots

One class, one semester, free for the school

Teacher ambassadors

Teachers who like VOC bring in colleagues

Case studies

Before/after retention data from each pilot

Partnerships

Exam-prep centres, NGOs, public programmes

We acquire customers on two tracks, starting with low-cost organic channels. For learners: short videos with daily memory tips — for example, why you forget words and how to stop it — on Instagram Reels, TikTok and YouTube Shorts; a Telegram channel with a word of the day and a quiz bot that leads into the app; a referral programme where both friends unlock extra drill packs; and a weekly "memory report" card that learners can share, which brings in their friends. For institutions: free one-semester pilots with a single class, so saying yes is easy; teacher ambassadors who recommend VOC to colleagues; case studies with before-and-after retention data from each pilot; and partnerships with exam-prep centres, education NGOs and public programmes. We will only spend on paid ads once we know which channels convert.

From product to a personal memory model

Product → Intelligence → Memory Twin

01 · NOW

Product phase

Build the core learning app.

- Vocabulary
- Grammar
- Personal forgetting model

02 · NEXT

Intelligence phase

Make the diagnosis smarter.

- Better diagnosis
- Institutional dashboard

03 · FUTURE

Memory Twin

A full model of how you learn, why you forget, how you recover.

- Predictive retention
- Cognitive mapping
- Personalised paths

The roadmap has three phases. Now, the product phase: the core app with vocabulary, grammar and the personal forgetting model. Next, the intelligence phase: a more precise diagnosis of why learners forget, and the institutional dashboard that schools and organisations pay for. In the future: the Memory Twin — a complete personal memory model that understands how you learn, why you forget, and how you recover. That enables predictive retention, a map of your cognitive strengths and weaknesses, and learning paths built entirely around you.

The people building VOC



Asadbek Abdukaxxarov

DESIGNER

UX/UI, visual design and
design systems.



Xolmirzayev Azimjon

PRODUCT MANAGER

Product strategy, learning systems
and roadmap.

DESIGN + PRODUCT + TECHNOLOGY

VOC is built by a two-person team. Asadbek Abdukaxxarov is the designer, responsible for UX/UI, visual design and the design system. Xolmirzayev Azimjon is the product manager, responsible for product strategy, the learning system and the roadmap. Together the team combines design, product and technology.

Who we will hire next, in order

Today we cover design and product. Next: the people who build the engine and sell it.

01 · FIRST

ML / data engineer

Builds and trains the memory model: stability, recall prediction, diagnosis.

Why: it is VOC's core technology.

02

Mobile / full-stack engineer

Ships the app on iOS, Android and web, plus the school dashboard.

Why: gets VOC into users' hands.

03

Growth & partnerships lead

Runs learner acquisition and closes pilots with schools and centres.

Why: turns pilots into paid licences.

04

Learning content lead

A linguist or teacher who designs word lists, grammar drills and tests.

Why: quality of what learners practise.

Advisor we are looking for: a researcher in learning science or cognitive psychology.

Our current team covers design and product. Our next hires, in priority order: First, an ML or data engineer, because the memory model — estimating stability, predicting recall and diagnosing mistakes — is VOC's core technology. Second, a mobile or full-stack engineer to ship the app on iOS, Android and web, plus the institutional dashboard. Third, a growth and partnerships lead who runs learner acquisition and turns free pilots with schools and learning centres into paid licences. Fourth, a learning content lead — a linguist or experienced teacher — who designs the word lists, grammar drills and tests. We are also looking for an advisor from learning science or cognitive psychology to validate the memory model.

VOC — PERSONAL MEMORY LAB

Memory is personal.

VOC doesn't just schedule your next review.

It diagnoses why you forget.

VOC Personal Memory Lab

azimjonxolmirzayev30@gmail.com

To sum up: memory is personal. Every learner forgets different words, at different speeds, for different reasons. Most tools only decide when to show you a word again. VOC goes further: it models your memory, predicts when you will forget, and diagnoses why — and then gives you the practice that fixes it. Thank you. Contact details are on the slide.